

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience building automated data quality and migration testing at Deutsche Bundesbank (AnaCredit), directly relevant to IT audit and data analytics work: validating business logic, detecting anomalies, and producing auditable evidence trails. Combines this with SQL/database skills and a structured, detail-oriented working style suited to complex, evidence-based technical review.

Core Competencies

- **IT Audit-Adjacent Data Quality Work:** Automated migration testing, error detection, and auditable evidence trails (Deutsche Bundesbank AnaCredit)
- **Data Analytics:** Python (Pandas, NumPy), SQL, SQLite, statistical anomaly detection
- **Business Process Understanding:** Regulatory reporting workflows and business-process context for banking data
- **Programming:** Python, SQL, Java, R, C++
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means
- **Cloud & Tools:** Azure Cognitive Services, AWS, Jira, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reported banking data, with automated Jira ticketing and reporting-bank notification
 - Significantly reduced response time to data quality issues through automated evidence generation

Projects

- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built a normalized SQLite database with foreign-key integrity checks for simulated bank transactions (150 accounts)
 - Staged data cleaning with a fully auditable log recording table, column, old/new value, action, and timestamp – designed to be audit-ready
 - Produced quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data; built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, numerical optimization.
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.